Surprisal-Advantage Gating for LLM Reasoning

Anonymous ACL submission

Abstract

Reinforcement learning (RL) has become the dominant paradigm for improving reasoning in large language models (LLMs), yet policy gradient methods suffer from *entropy collapse* and *self-reinforcing bias*: the policy prematurely narrows its reasoning strategies while over-allocating gradient to problems it already solves. We show that surprisal-advantage gating, which weights each gradient term by a sigmoid of the product of advantage and action surprisal, addresses both pathologies through two complementary mechanisms: (i) the gate discards gradient from terms that drive entropy collapse, preserving the policy's exploration capacity; (ii) the gate asymmetrically preserves gradient from surprising correct solutions while suppressing gradient from surprising failures, reallocating gradient budget toward harder problems. We integrate this gating into GRPO and SDPO, designing sequence-level and token-level variants. On mathematical reasoning benchmarks (AIME 2024/2025, HMMT 2025, AMO-Bench), GRPO-DG consistently outperforms GRPO across all pass@k values, with the gap growing with k (up to $+16.7\%$ on AIME 2024), and outperforms other exploration-preserving baselines. A difficulty-stratified analysis confirms that gains concentrate on problems the baseline cannot solve ($+45\%$ on the hardest stratum), while easy problems are unaffected. On scientific reasoning (SciKnowEval), SDPO-DG improves over SDPO by $+6.1\%$ on pass@16 across four domains.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has become the dominant paradigm for enhancing reasoning in large language models (OpenAI, 2024; DeepSeek-AI, 2025). Methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) use outcome-based rewards with group-relative baselines, while more recent on-policy distillation methods like SDPO (Hübotter et al., 2026; Zhao et al., 2026) construct per-token signals based on teacher-student divergence. Both families of methods have advanced LLM reasoning significantly, yet they share two failure modes that limit their effectiveness.

The first failure mode is **entropy collapse**: the policy rapidly concentrates on a narrow set of reasoning paths, losing the ability to discover alternative strategies (Cui et al., 2025). This manifests as a growing discrepancy between pass@1 (average accuracy) and pass@k for large k (solution coverage): the model improves at producing a few confident strategies but fails to explore new ones (Yue et al., 2025). Entropy collapse limits the effectiveness of test-time compute scaling (Snell et al., 2024; OpenAI, 2024), since generating additional samples merely reproduces previously found solutions rather than discovering new reasoning paths.

The second failure mode is **self-reinforcing bias**: policy gradient inherently over-weights easy problems at the expense of hard ones (Osband, 2026). In a simplified setting, Osband (2026) show that, unlike supervised learning, which assigns uniform gradient weight to all training examples, policy gradient weights each problem by the probability that the policy selects the correct solution. This probability is high for easy problems and low for hard ones, so policy gradient under-weights precisely hard problems where the policy still needs learning signals (see Appendix D for a formal derivation).

Recent works attempt to address entropy collapse. Cui et al. (2025) propose CovClip, which detaches tokens whose covariance between log-probability and advantage exceeds a threshold, directly targeting the entropy-collapsing mechanism. Cheng et al. (2025) augment the advantage with an entropy bonus to encourage exploration. While both improve over standard GRPO, neither method addresses self-reinforcing bias.

In this work, we show that *surprisal-advantage*

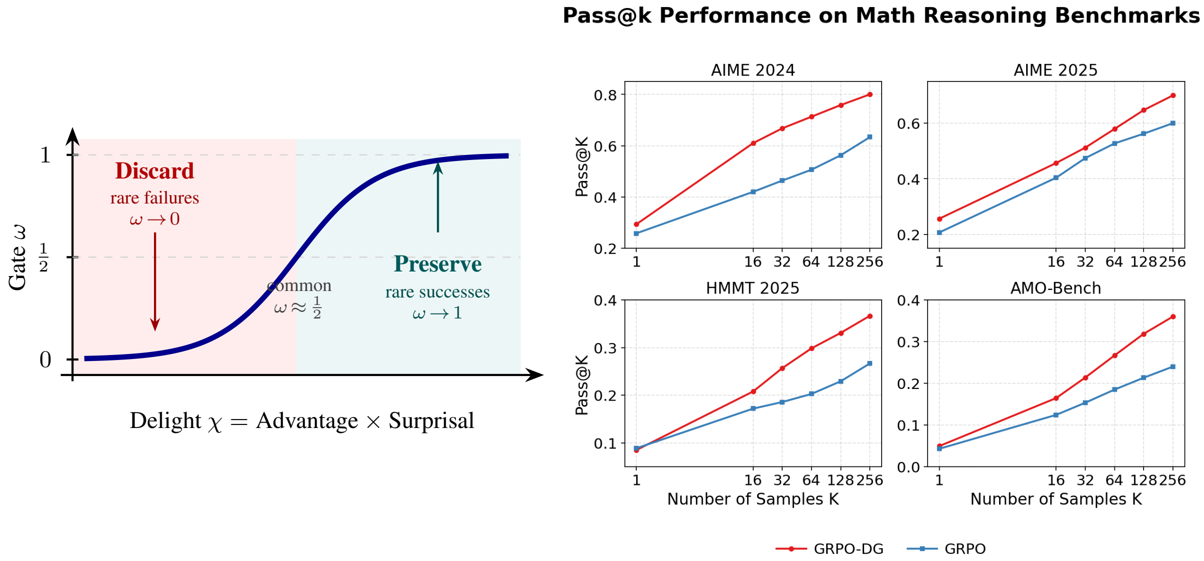

Figure 1: **Left: The DG gate mechanism.** The gate $\omega = \sigma(\chi/\eta)$ weights each gradient term by a sigmoid of delight (advantage × surprisal). Rare actions with negative advantage (entropy-collapsing terms) are discarded; rare actions with positive advantage (novel successes) are preserved; common actions receive standard gradient. **Right: Effect on pass@k.** GRPO-DG (red) consistently outperforms GRPO (blue) across all k, with the gap growing monotonically (up to $+16.7\%$ on AIME 2024 at k=256). The growing gap indicates that the gated policy explores a broader set of reasoning strategies. A difficulty-stratified analysis (Section 4.3) further shows that gains concentrate on the hardest problems ($+45\%$), while easy problems are unaffected.

gating, introduced by Osband (2026) as the Delightful Gradient (DG), simultaneously addresses both pathologies through a single, computationally lightweight mechanism (Figure 1, right). The gate weights each policy gradient term by a sigmoid of *delight*, the product of advantage and action surprisal. We provide new analysis explaining why this is effective for LLM reasoning, extending the original DG analysis (which assumes a bandit setting and validates only on simple tasks like MNIST) to the more complex LLM reasoning settings. We identify two complementary mechanisms: (1) **entropy preservation**, where the gate discards gradient from the terms that drive entropy collapse; and (2) **gradient budget reallocation**, where the gate's asymmetric treatment of surprising successes vs. surprising failures implicitly redirects gradient toward harder problems. These mechanisms are complementary: entropy preservation maintains the policy's capacity to explore diverse strategies, while gradient budget reallocation focuses that exploration where it matters most.

Our contributions are as follows:

- We integrate DG with GRPO and SDPO, designing sequence-level and token-level gating variants respectively.
- We provide novel analysis connecting DG to entropy dynamics and gradient budget reallocation in LLM reasoning.
- We validate empirically on mathematical reasoning (up to $+16.7\%$ at pass@256, with gains growing with k and concentrating on the hardest problems) and scientific reasoning ($+6.1\%$ average pass@16 across four domains).

2 Related Work

RL for LLM reasoning. The application of RL to enhance LLM reasoning was catalyzed by OpenAI's o1 (OpenAI, 2024) and DeepSeek-R1 (DeepSeek-AI, 2025), which demonstrated that pure RL training can elicit sophisticated reasoning behaviors without human-annotated demonstrations. GRPO (Shao et al., 2024) simplified PPO (Schulman et al., 2017) by replacing the learned value function with group-relative advantage estimation, substantially reducing compute and memory requirements. Liu et al. (2025); Yu et al. (2025) subsequently identified optimization biases in GRPO (difficulty bias from standard-deviation normalization and length bias from per-sequence normalization) and proposed debiased corrections. SDPO and OPSD (Hübotter et al., 2026; Agarwal et al., 2026; Zhao et al., 2026) addressed the credit-assignment bottleneck by constructing a self-teacher policy conditioned on privileged information, enabling dense per-token learning signals that provide more informative gradient than outcome-level rewards.

Exploration in RL for LLMs. Maintaining exploration is critical for reasoning tasks where the solution space is vast. Classical entropy bonuses (Haarnoja et al., 2018) uniformly encourage stochasticity without distinguishing productive from unproductive randomness. Cui et al. (2025) established that entropy reduction under policy gradient is governed by the covariance between log-policy and advantage, and proposed CovClip, which detaches tokens whose pointwise covariance exceeds a threshold. Cheng et al. (2025) take a complementary approach, augmenting the advantage with an entropy-based term that rewards exploratory behaviors.

We show in Section 3.3 that CovClip's detachment criterion and DG's gating mechanism both target the same quantity: under zero-mean advantages, the tokens CovClip detaches correspond to those with high negative delight, which are precisely the entropy-collapsing terms. This connection, which we establish through the covariance-delight identity (Eq. 10), reveals that DG provides a smooth, sigmoid-weighted alternative to CovClip's binary detachment, while additionally amplifying positive-delight terms (surprising successes).

Delightful Policy Gradient. Osband (2026) introduced DG, gating each policy gradient term by a sigmoid of delight (the product of advantage and action surprisal). The theoretical analysis proves variance reduction and gradient budget reallocation in K-armed bandits, which are validated on simple tasks like MNIST classification and digit reversal. We extend this to LLM reasoning: providing new analysis connecting DG to entropy dynamics (Section 3.3), integrating it with LLM RL methods GRPO and SDPO (Section 3.2), and validating the budget reallocation effect through difficulty-stratified empirical analysis on challenging mathematical reasoning benchmarks (Section 4).

3 Methods

Notation. We consider a training set $\mathcal{D} = \{(x, y^*)\}$ where x is a prompt or question and y^* is the ground-truth answer. For each prompt x, an LLM generates a response $y = \{y_1, \ldots, y_T\}$ consisting of T tokens, sampled from the policy $y \sim \pi_\theta(\cdot|x)$, where θ denotes the model parameters. A reward function $r(y^*, y) \in \{0, 1\}$ assigns 1 if y is equivalent to y^*, and 0 otherwise.

3.1 Background

GRPO. Group Relative Policy Optimization (Shao et al., 2024) estimates the advantage baseline from a group of G responses for a given prompt, $\{y_i\}_{i=1}^G \sim \pi_\theta(\cdot \mid x)$, rather than from a learned value function. This simplifies the algorithm and substantially reduces compute and memory requirements. Following Liu et al. (2025), we adopt the debiased variant that removes standard deviation normalization (which creates difficulty bias) and replaces per-sequence length normalization with global token normalization (which removes length bias):

$$\mathcal{J}^{\mathrm{GRPO}}(\theta) = \mathbb{E}_{x,\{y_i\}_{i=1}^G}\left[\frac{1}{\sum_i |y_i|}\sum_{i=1}^{G}\sum_{t=1}^{|y_i|} L_{i,t}\right], \tag{1}$$

$$L_{i,t} = \min\bigl(r_{i,t}\, A_{i,t},\ \bar{r}_{i,t}\, A_{i,t}\bigr),$$

where the expectation is over $x \sim \mathcal{D}$ and $\{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot \mid x)$, $r_{i,t} = \pi_\theta(y_{i,t} \mid x, y_{i,<t})/\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})$ is the per-token importance ratio, $\bar{r}_{i,t} = \mathrm{clip}(r_{i,t},\, 1-\epsilon,\, 1+\epsilon)$, and $A_{i,t} = r(y^*, y_i) - \frac{1}{G}\sum_{j=1}^{G} r(y^*, y_j)$.

SDPO. Self-Distillation Policy Optimization (Hübotter et al., 2026; Zhao et al., 2026) addresses GRPO's credit-assignment bottleneck by constructing a self-teacher policy conditioned on privileged information f (e.g., correct rollouts, reference solutions, environment feedback). The loss is based on the Jensen–Shannon divergence between teacher and student distributions:

$$\mathcal{L}^{\mathrm{SDPO}}(\theta) = \mathbb{E}_{x,y}\left[\frac{1}{|y|}\sum_{t=1}^{|y|} \mathrm{JSD}_\alpha\bigl(\pi_t^{\text{teach}},\ \pi_{\theta t}\bigr)\right], \tag{2}$$

where $\pi_t^{\text{teach}} = \mathrm{sg}(\pi_\theta(y_t|x, y_{<t}, f))$ is the stop-gradient teacher, $\pi_{\theta t} = \pi_\theta(y_t|x, y_{<t})$ is the student, and $\mathrm{JSD}_\alpha(p, q) = \alpha\,\mathrm{KL}(p\|m) + (1-\alpha)\,\mathrm{KL}(q\|m)$ with $m = \alpha p + (1-\alpha) q$. To integrate with DG, we reformulate the loss as an equivalent policy gradient surrogate with per-token, per-vocabulary

advantages:

$$\mathcal{J}^{\text{SDPO}} = \mathbb{E}_{x,y}\left[\frac{1}{|y|}\sum_{t}\sum_{v} A_{t,v}\log\pi_\theta(v|x,y_{<t})\right], \tag{3}$$

where the per-token, per-vocabulary advantage is:

$$A_{t,v} = \text{sg}\left(\log\frac{\alpha\pi_f(v) + (1-\alpha)\pi_\theta(v|x,y_{<t})}{\pi_\theta(v|x,y_{<t})}\right),$$

with $\pi_f(v) = \pi_\theta(v|x, y_{<t}, f)$ denoting the teacher distribution. The per-token, per-vocabulary structure of these advantages enables fine-grained gating when we integrate DG with SDPO.

3.2 Delightful Gradient for LLM Reasoning

The DG gate. At timestep t, given an action y_t sampled from the policy $\pi_\theta(\cdot \mid x, y_{<t})$ with advantage A_t, we define the *action surprisal* as $l_t = -\log\pi_\theta(y_t \mid x, y_{<t})$ and the *delight* as the product $\chi_t = A_t \cdot l_t$. The DG gate (Figure 1, left) is then:

$$\omega_t = \sigma\left(\frac{\chi_t}{\eta}\right), \tag{4}$$

where σ is the sigmoid function and $\eta > 0$ is a temperature parameter controlling sharpness. The gate's behavior depends on the interaction between advantage sign and action rarity: when a rare action (l_t is large) has positive advantage, the delight is large and the gate opens ($\omega_t \approx 1$); when a rare action has negative advantage, the delight is strongly negative and the gate closes ($\omega_t \approx 0$); for actions the policy already assigns high probability (l_t is small), the delight is near zero regardless of advantage sign, and the gate stays near $1/2$, approximately recovering standard gradient behavior.

GRPO-DG. In GRPO, advantages are defined at the sequence level. We therefore define the gate using the sequence-level geometric-mean surprisal, which averages token-level log-probabilities across the response:

$$\begin{aligned}\chi_i &= A_i \cdot \left(-\tfrac{1}{|y_i|}\log\pi_\theta(y_i \mid x)\right), \\ \omega_i &= \sigma\left(\chi_i/\eta\right),\end{aligned} \tag{5}$$

$$\mathcal{J}^{\text{GRPO-DG}} = \mathbb{E}_{x,\{y_i\}}\left[\frac{1}{\sum_i |y_i|}\sum_{i=1}^{G}\omega_i\sum_{t=1}^{|y_i|}L_{i,t}\right]. \tag{6}$$

The gate ω_i operates at the sequence level: novel successes ($\chi_i \gg 0$) receive full gradient weight, novel failures ($\chi_i \ll 0$) are suppressed (protecting the policy's willingness to explore uncommon strategies), and familiar responses ($|\chi_i| \approx 0$) receive standard half-weight.

SDPO-DG. Given the policy gradient reformulation, SDPO provides per-token, per-vocabulary advantages, enabling fine-grained gating at the token level.

$$\begin{aligned}\chi_{t,v} &= A_{t,v} \cdot \left(-\log\pi_\theta(v \mid x, y_{<t})\right), \qquad (7)\\ \omega_{t,v} &= \sigma(\chi_{t,v}/\eta),\end{aligned}$$

$$\mathcal{J}^{\text{SDPO-DG}} = \mathbb{E}_{x,y}\left[\frac{1}{|y|}\sum_{t}\sum_{v}\omega_{t,v}A_{t,v}\log\pi_\theta^t(v)\right], \tag{8}$$

where $\pi_\theta^t(v) = \pi_\theta(v|x, y_{<t})$. Here the gate operates at the finest possible granularity: each vocabulary token at each position receives its own weight based on its local delight. A token v receives high weight ($\omega_{t,v} \to 1$) when the self-teacher assigns it higher probability than the student (positive $A_{t,v}$) *and* the student currently assigns it low probability (high surprisal). This is exactly the case where the privileged teacher has discovered a promising token that the student has not yet learned. Conversely, tokens that both teacher and student assign low probability are attenuated ($\omega_{t,v} \to 0$), preventing the student from being punished for tokens carrying no useful signal.

3.3 Why Does the DG Gate Work?

Having defined the method, we now analyze *why* the delight gate is effective for LLM reasoning. We identify two complementary mechanisms.

Mechanism 1: Entropy preservation via delight. We connect the DG gate to the mechanism governing entropy dynamics under policy gradient. Cui et al. (2025) establish that under natural policy gradient (which shares the KL-constrained geometry of PPO and GRPO), the per-step entropy change of the policy at a given state is:[1]

$$\Delta H \approx -\eta \cdot \text{Cov}_{a\sim\pi_\theta}\left(\log\pi_\theta(a \mid s),\ A(s,a)\right). \tag{9}$$

The advantage $A(s,a) = Q(s,a) - V(s)$ is zero-mean in expectation by definition. Given $\mathbb{E}_\pi[A] = 0$, the centering term in the covariance vanishes,

[1]We use a to denote an action and s to denote a state, following the notation in the RL literature.

yielding a direct connection to delight:

$$\begin{aligned}\mathrm{Cov}(\log \pi, A) &= \mathbb{E}_{a\sim\pi}[\log \pi(a)\cdot A(a)] \\ &= -\mathbb{E}_{a\sim\pi}\big[\underbrace{A(a)\cdot(-\log \pi(a))}_{\chi(a)\text{ (delight)}}\big].\end{aligned} \tag{10}$$

Substituting into Eq. 9, we obtain:

$$\Delta H \approx \eta\cdot \mathbb{E}_{a\sim\pi}[\chi(a)]. \tag{11}$$

Entropy decreases when expected delight is negative, and increases when it is positive. The DG gate $\omega = \sigma(\chi/\eta)$ directly intervenes on this mechanism. Terms with $\chi \ll 0$ (rare actions with negative advantage) would push $\mathbb{E}[\chi]$ more negative and accelerate entropy collapse; DG discards their gradient by setting $\omega \to 0$. Terms with $\chi \gg 0$ (rare actions with positive advantage) contribute positive delight that counteracts collapse; DG preserves them with $\omega \to 1$. Common actions ($|\chi| \approx 0$) are treated neutrally with $\omega \approx 1/2$. By selectively attenuating negative-delight terms and preserving positive-delight terms, the gate shifts the effective $\mathbb{E}[\chi]$ upward, counteracting entropy collapse without uniformly encouraging stochasticity.

Mechanism 2: Gradient budget reallocation. Entropy preservation explains why DG maintains the policy's *capacity* to explore. We now show that the gate also focuses exploration where it matters most.

The key insight is a distributional asymmetry in surprisal across problem difficulties. Let $p_x = \frac{1}{G}\sum_{i=1}^{G} r_i$ denote the empirical success rate for prompt x within a group. On hard problems ($p_x \ll 1$), correct responses tend to have higher surprisal because the policy has had fewer opportunities to reinforce them. On easy problems ($p_x \approx 1$), incorrect responses tend to have higher surprisal because they go against the policy's learned preference. We verify this asymmetry empirically in Section 4.3.

Combined with GRPO's mean-centered advantages, this asymmetry produces a systematic pattern in gate values. On hard problems, correct responses have large positive advantage (the mean reward is low) *and* high surprisal, yielding $\chi \gg 0$ and $\omega \to 1$: their reinforcement signal is fully preserved. Incorrect responses have near-zero advantage, yielding near-neutral gating. On easy problems, the pattern reverses: incorrect responses have large negative advantage (the mean reward is high) and high surprisal ($\chi \ll 0, \omega \to 0$, punishment suppressed), while correct responses have near-zero advantage (treated neutrally). The net effect is that DG preserves the signal for novel correct solutions (concentrating on hard problems) while discarding the signal for rare failures (concentrating on easy problems), implicitly reallocating gradient budget toward harder problems when the gradients are averaged across a batch of easy and hard problems.

This differs from the analysis of Osband (2026), which identifies $\pi(y^*)$ as a direct difficulty proxy in the bandit setting. In LLM reasoning, there is no single correct response, so surprisal is not a direct proxy for difficulty but an indirect one mediated by the distributional asymmetry above. The mechanism selectively amplifies learning from *surprising successes* and ignores *surprising failures*, which correlates with difficulty-based gradient weight reallocation.

The two mechanisms are complementary: entropy preservation maintains the policy's ability to generate diverse candidates, while gradient budget reallocation ensures that when a surprising candidate succeeds on a hard problem, its learning signal is not diluted. We validate both predictions in Section 4.3.

3.4 Ablation: Difficulty-Aware Gating

To isolate the contribution of entropy preservation (Mechanism 1) from budget reallocation (Mechanism 2), we introduce an ablation called DiffGate that implements budget reallocation through an explicit problem-level difficulty signal rather than response-level delight:

$$\mathcal{J}^{\text{DiffGate}}(\theta) = \mathbb{E}_{x,\{y_i\}}\left[\frac{\omega_x}{\sum_i |y_i|}\sum_{i=1}^{G}\sum_{t=1}^{|y_i|} L_{i,t}\right], \tag{12}$$

where $\omega_x = \sigma((d_x - 0.5)/\eta_d)$ and $d_x = 1 - \frac{1}{G}\sum_{i=1}^{G} r(y^*, y_i)$ is the group failure rate. This is a sigmoid gate on problem difficulty: it monotonically upweights harder problems ($\omega_x \to 1$ as $d_x \to 1$) and downweights easy problems ($\omega_x \to 0$ as $d_x \to 0$). Unlike GRPO-DG, DiffGate does not operate at the response level and does not differentially treat surprising versus familiar responses, so it does not benefit from entropy preservation. If GRPO-DG outperforms DiffGate, the additional gain can be attributed to entropy preservation.

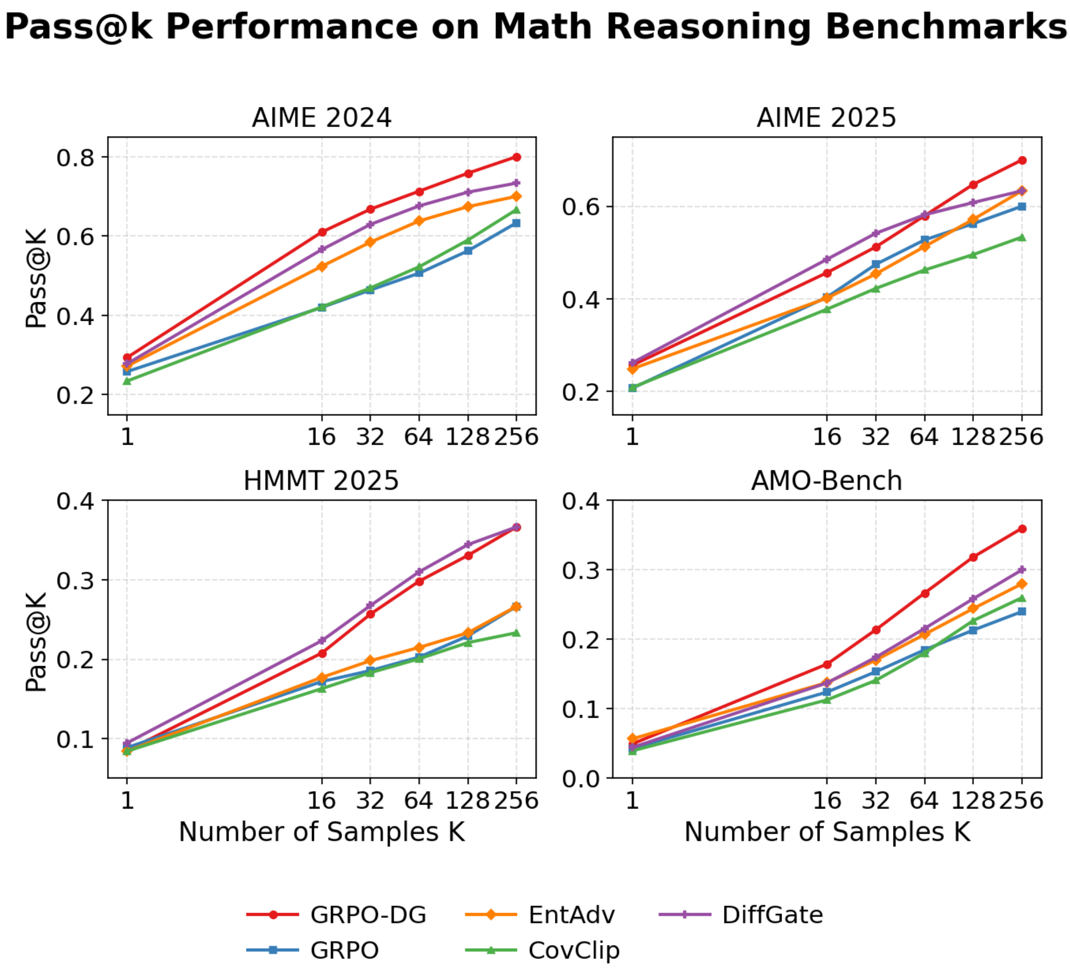

Figure 2: Pass@k curves on four mathematical reasoning benchmarks (Qwen3-4B-Base). GRPO-DG (red) consistently outperforms GRPO (blue) across all k values, with the gap growing monotonically (up to $+16.7\%$ on AIME 2024 at k=256). CovClip and EntAdv also improve over GRPO and are competitive in some settings, but GRPO-DG leads overall across benchmarks and k values. DiffGate (budget reallocation only) performs strongly but underpeforms GRPO-DG overall, confirming that entropy preservation provides additional value.

4 Experiments

4.1 Mathematical Reasoning with GRPO-DG

Setup. We train Qwen3-4B-Base (Yang et al., 2025) using GRPO-DG on the DAPO math dataset (Yu et al., 2025), which provides a curated collection of mathematical problems with verifiable answers. We compare against four methods: (1) **GRPO** (Shao et al., 2024; Liu et al., 2025), the debiased variant described in Eq. 1; (2) **EntAdv** (Cheng et al., 2025), which augments the advantage with an entropy-based term to promote exploration; (3) **CovClip** (Cui et al., 2025), which detaches tokens whose covariance between log-policy and advantage exceeds a threshold; and (4) **DiffGate**, our ablation that gates by problem-level difficulty (Eq. 12). All methods share the same base model, training data, and hyperparameters except for method-specific parameters (see Appendix E for full details). Each model is trained on a single node of 8$\times$H200 GPUs for approximately 72 hours.

Evaluation. We evaluate on four challenging mathematical reasoning benchmarks: AIME 2024, AIME 2025, HMMT 2025, and AMO-Bench. We report pass@k for $k \in \{1, 16, 32, 64, 128, 256\}$, estimated unbiasedly from 256 samples per problem.

Results. Figure 2 summarizes the results (exact numbers in Table 2, Appendix). GRPO-DG achieves the best or tied-best pass@256 on every benchmark while maintaining competitive pass@1, indicating that the gate improves solution coverage without sacrificing average accuracy.

The most striking pattern is the growing gap

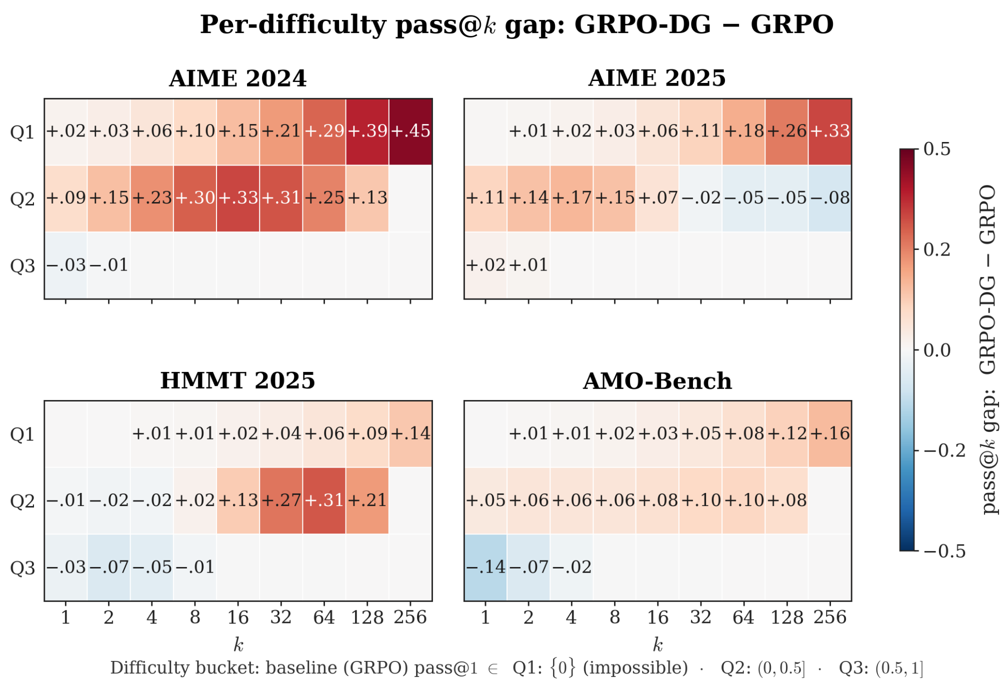

Figure 3: Per-problem pass@k gap (Δ_k = pass@k(GRPO-DG) − pass@k(GRPO)) stratified by baseline difficulty across four benchmarks. Rows are difficulty buckets based on the GRPO baseline's pass@1: Q1 (impossible: pass@1=0), Q2 (medium: $(0, 0.5]$), Q3 (easy: $(0.5, 1]$). Red indicates GRPO-DG outperforms GRPO. Gains concentrate on the hardest stratum (Q1) and grow monotonically with k, reaching $+45\%$ on AIME 2024. Easy problems (Q3) show no degradation.

between GRPO-DG and GRPO as k increases. On AIME 2024, the gap is $+3.5\%$ at pass@1 but widens to $+16.7\%$ at pass@256 (80.0% vs. 63.3%). Similar scaling appears on AIME 2025 ($+5.0\%$ at pass@1, $+10.0\%$ at pass@256), HMMT 2025 ($+10.0\%$ at pass@256), and AMO-Bench ($+12.0\%$ at pass@256). This monotonically growing gap is the signature of improved exploration: the gated policy covers a broader set of reasoning strategies, so additional samples surface genuinely new valid solutions rather than duplicating previously found ones.

CovClip and EntAdv both improve substantially over GRPO, confirming that addressing entropy collapse is valuable. CovClip ties GRPO-DG at pass@256 on AIME 2024, but falls short on the other three benchmarks. DiffGate, which implements only budget reallocation without entropy preservation, also performs strongly, particularly on HMMT 2025. However, it trails GRPO-DG on AIME 2024 and AMO-Bench (visible in Figure 2), confirming that entropy preservation provides value beyond budget reallocation alone.

Table 1: SciKnowEval results (%) with Llama-3-8B. DG improves both GRPO and SDPO across all domains. Best per column in **bold**.

	Pass@1				*Pass@16*			
	Bio.	Chem.	Phys.	Mat.	Bio.	Chem.	Phys.	Mat.
GRPO	53.7	73.6	68.5	74.2	64.7	84.3	79.2	80.5
GRPO-DG	56.9	77.5	70.3	77.0	69.9	87.5	82.9	85.3
SDPO	56.8	80.9	73.7	75.3	67.8	85.7	85.6	79.7
SDPO-DG	**65.1**	**81.0**	**79.2**	**77.8**	**78.3**	**90.0**	**89.0**	**85.8**

4.2 Scientific Reasoning with SDPO-DG

To test generalization across model families and task domains, we apply DG to both GRPO and SDPO using Llama-3-8B (a different model family and scale from the math experiments). See Appendix E for training hyperparameters. We evaluate on SciKnowEval (Feng et al., 2024), a benchmark spanning four scientific subjects: biology, chemistry, physics, and materials science. Since SciKnowEval consists of multiple-choice problems, pass@k at large k becomes less informative, so we report up to k=16.

Table 1 reveals three findings. First, DG consistently improves both base algorithms: GRPO-DG outperforms GRPO and SDPO-DG outperforms

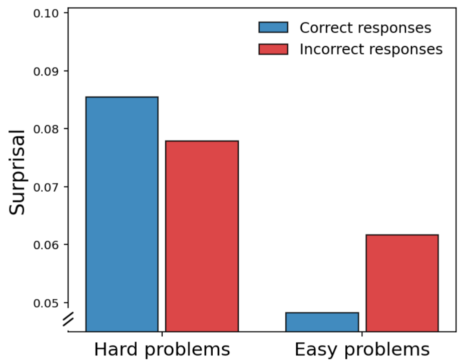

Figure 4: Empirical validation of the surprisal asymmetry underlying Mechanism 2. On hard problems (low group accuracy), correct responses have higher surprisal than incorrect ones. On easy problems, the pattern reverses.

SDPO across all domains. Second, SDPO-DG achieves the best results overall, suggesting that the combination of dense credit assignment (from SDPO) and exploration-preserving gating (from DG) is particularly effective. Third, improvements are larger at pass@16 than at pass@1 (averaging $+6.1$ vs. $+4.4$ over SDPO), echoing the broader-coverage signature observed in the math experiments. These results confirm that DG generalizes across model families (Qwen to Llama) and task domains (mathematics to science).

4.3 Analysis

Surprisal asymmetry. The budget reallocation mechanism relies on a distributional assumption: correct responses on hard problems should have higher surprisal than incorrect ones, while the reverse should hold on easy problems. Figure 4 validates this empirically using data from GRPO training. On hard problems, correct responses exhibit substantially higher surprisal because the policy has not yet learned to produce these solutions fluently. On easy problems, incorrect responses have higher surprisal because they go against the learned preference. This asymmetry enables the DG gate to achieve difficulty-based budget reallocation through response-level delight.

Difficulty-stratified analysis. To directly test the budget reallocation prediction, we stratify problems by the GRPO baseline's pass@1 into three buckets: Q1 (impossible: pass@1=0), Q2 (medium: $(0, 0.5]$), Q3 (easy: $(0.5, 1]$). Figure 3 shows the pass@k gap between GRPO-DG and GRPO for each bucket. On the hardest problems (Q1), the gap grows monotonically with k, reaching $+45\%$ on AIME 2024, $+33\%$ on AIME 2025, $+16\%$ on AMO-Bench, and $+14\%$ on HMMT 2025 at k=256. Medium problems (Q2) show positive gaps peaking at intermediate k. Easy problems (Q3) are mostly flat, showing no degradation. This confirms the gradient budget reallocation mechanism: DG concentrates learning on harder problems while leaving easy problems unaffected.

Entropy dynamics. To test the entropy preservation prediction, we track policy entropy throughout training (Figure 5, Appendix). GRPO-DG stabilizes at $\sim$0.13, substantially higher than GRPO at $\sim$0.05. CovClip maintains even higher entropy but with large variance, consistent with its binary detachment mechanism. In SDPO training, the effect is more pronounced: SDPO-DG reaches $\sim$3.5 while SDPO collapses below 1.5. These patterns confirm that selectively discarding negative-delight terms counteracts entropy collapse as predicted.

Ablation analysis. DiffGate implements gradient budget reallocation without entropy preservation. As shown in Figure 2, DiffGate performs strongly: it matches or exceeds GRPO-DG on pass@1 for some benchmarks, indicating that budget reallocation alone accounts for much of the gain at low k. However, at intermediate and high k (k = 16–256), GRPO-DG consistently leads, with the gap most visible on AIME 2024 and AMO-Bench. This confirms that entropy preservation provides additional diversity that only surfaces when sampling more candidates.

5 Conclusion

We integrated the Delightful Gradient gate into GRPO and SDPO for LLM reasoning, and provided new analysis connecting the gate to entropy dynamics and gradient budget reallocation. The gate addresses two failure modes of policy gradient methods, *entropy collapse* and *self-reinforcing bias*, through a single mechanism that discards gradient from negative-delight terms (counteracting entropy collapse) while preserving gradient from positive-delight terms (concentrating learning on hard problems). Empirically, GRPO-DG consistently outperforms GRPO with gains growing with k and concentrating on the hardest problems, while SDPO-DG improves across four scientific domains. The method generalizes across model families (Qwen, Llama) and task domains (mathematics, science).

Limitations

Our work has several limitations. GRPO-DG operates at the sequence level, which may miss token-level structure (SDPO-DG addresses this at the cost of requiring a self-teacher). While we validate on models up to 8B parameters, a comprehensive scaling study is left for future work.

Potential Risks

Our method improves the reasoning capabilities of language models, which broadly carries the same risks as other advances in LLM reasoning: models that reason more effectively could be misused to generate persuasive misinformation or assist with harmful tasks. However, our contribution is a training-time optimization technique that does not introduce new capabilities beyond what the base model and training data already enable. The method does not involve any human subjects or private data.

Use of AI Assistants

LLM-based writing assistants were used for language editing and formatting. All research contributions, including theoretical analysis, experimental design, implementation, and interpretation of results, are entirely the authors' own work.

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A Detailed Numerical Results

Table 2 reports the exact pass@1 and pass@256 values for all methods on the four math reasoning benchmarks.

Table 2: Pass@1 and Pass@256 (%) on four math reasoning benchmarks. Best in **bold**, second best $\underline{\text{underlined}}$.

	AIME24		AIME25		HMMT25		AMO-Bench	
	@1	@256	@1	@256	@1	@256	@1	@256
GRPO-DG	**29.4**	**80.0**	$\underline{25.7}$	**70.0**	8.5	**36.7**	$\underline{5.0}$	**36.0**
GRPO	25.8	63.3	20.7	60.0	$\underline{8.9}$	26.7	4.3	24.0
EntAdv	27.2	70.0	24.9	63.3	8.4	26.7	**5.7**	28.0
CovClip	$\underline{27.6}$	**80.0**	23.3	$\underline{66.7}$	8.7	$\underline{30.0}$	4.5	$\underline{32.0}$
DiffGate	27.8	$\underline{73.3}$	**26.2**	63.3	**9.5**	**36.7**	4.5	30.0

B Additional Ablation: Positive-Only Gating

We additionally compare against a PosOnly ablation that restricts the GRPO update to correct responses only, with the DG gate applied:

$$\mathcal{J}^{\text{PosOnly}}(\theta) = \mathbb{E}_{x,\{y_i\}}\left[\frac{1}{\sum_i |y_i|}\sum_{i:\, r_i=1} \omega_i \sum_{t=1}^{|y_i|} L_{i,t}\right], \tag{13}$$

with $\omega_i = \sigma(\chi_i/\eta)$ as in Eq. 5. Since only correct responses ($r_i = 1$) contribute, the surprisal in χ_i measures how novel a correct solution is to the policy, which serves as a direct difficulty proxy (closer to the original DG analysis). Table 3 shows that PosOnly generally trails both DiffGate and GRPO-DG, particularly at mid-k values ($k = 16$–128), indicating that retaining negative-advantage gradient signal for common (non-surprising) failures provides useful learning signal that should not be discarded entirely.

Table 3: PosOnly ablation results: Pass@1 and Pass@256 (%).

	AIME24		AIME25		HMMT25		AMO-Bench	
	@1	@256	@1	@256	@1	@256	@1	@256
GRPO-DG	29.4	80.0	25.7	70.0	8.5	36.7	5.0	36.0
DiffGate	27.8	73.3	26.2	63.3	9.5	36.7	4.5	30.0
PosOnly	25.0	73.3	21.8	60.0	8.1	30.0	4.2	30.0

C Entropy Dynamics

To test the entropy preservation prediction, we track policy entropy throughout training (Figure 5). GRPO-DG stabilizes at $\sim$0.13, substantially higher than GRPO at $\sim$0.05. CovClip maintains even higher entropy but with large variance, consistent with its binary detachment mechanism. In SDPO training, the effect is more pronounced: SDPO-DG reaches $\sim$3.5 while SDPO collapses below 1.5. These patterns confirm that selectively discarding negative-delight terms counteracts entropy collapse as predicted.

D Self-Reinforcing Bias: Formal Derivation

We formalize the self-reinforcing bias of policy gradient following Osband (2026). Consider state s, correct action a^*, and binary reward with zero baseline. The expected gradients under supervised learning (SL), policy gradient (PG), and Delightful Policy Gradient (DG) are:

$$\text{SL:} \quad \nabla_\theta \log \pi_\theta(a^*|s), \tag{14}$$

$$\text{PG:} \quad \pi_\theta(a^*|s) \cdot \nabla_\theta \log \pi_\theta(a^*|s), \tag{15}$$

$$\text{DG:} \quad \pi_\theta(a^*|s) \cdot \sigma\big(-\log \pi_\theta(a^*|s)\big) \cdot \nabla_\theta \log \pi_\theta(a^*|s). \tag{16}$$

PG weights by $\pi_\theta(a^*|s)$, which is large for easy problems and small for hard ones. DG adds a corrective factor $\sigma(-\log \pi_\theta(a^*|s))$ that is larger for hard problems (low $\pi_\theta(a^*|s)$) and smaller for easy ones, partially canceling the bias and redistributing gradient toward harder problems. In LLM reasoning the setting is more complex (multiple correct responses, group-centered advantages), which we analyze in Section 3.3.

E Experimental Details

All methods share the same base model (Qwen3-4B-Base for math experiments, Llama-3-8B for science experiments), training data, and hyperparameters except for method-specific parameters. Each model is trained on a single node of 8$\times$H200 GPUs for approximately 72 hours. For pass@k estimation, we use the unbiased estimator from 256 samples per problem with a 12k token context window. The DG temperature parameter η is set to 1.0 for all experiments.

Dataset statistics. For math reasoning, we train on the DAPO math dataset, which contains approximately 17K problems with verifiable answers.

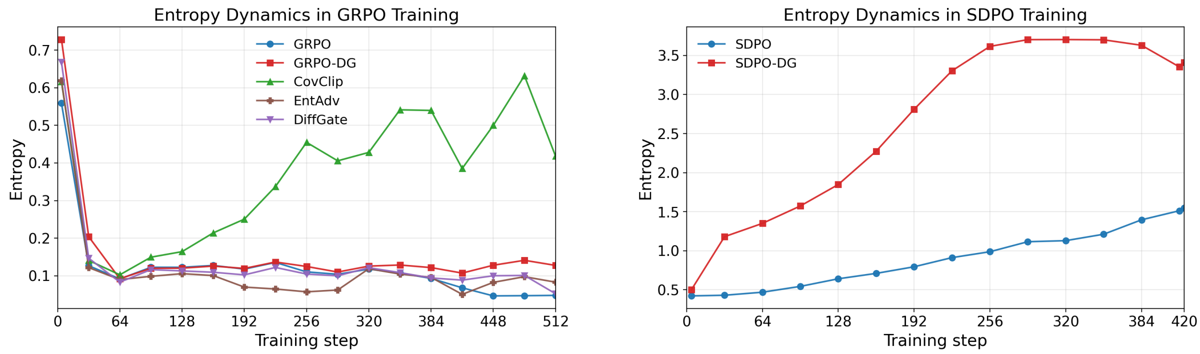

Figure 5: Policy entropy throughout training. **Left:** GRPO variants. All methods experience rapid initial entropy drop, but GRPO-DG stabilizes at a higher level ($\sim$0.13) than GRPO ($\sim$0.05). CovClip maintains even higher entropy but with large variance. **Right:** SDPO variants. SDPO-DG maintains substantially higher entropy than SDPO throughout training.

We evaluate on AIME 2024 (30 problems), AIME 2025 (30 problems), HMMT 2025 (30 problems), and AMO-Bench (50 problems). For scientific reasoning, we train and evaluate on SciKnowEval, which contains 2,162 multiple-choice questions split across four domains: biology (540), chemistry (540), physics (541), and materials science (541). For SciKnowEval, the train/test splits follow those used in the SDPO paper (Hübotter et al., 2026). All other datasets are used as provided by their authors without modification.

Table 4 lists the full set of hyperparameters used for GRPO training. Table 5 lists the hyperparameters used for the scientific reasoning experiments with SDPO.

F Licenses and Intended Use

All artifacts used in this work are publicly available under permissive licenses. The DAPO dataset and Qwen3-4B-Base model are released under the Apache 2.0 license. Llama-3-8B is released under the Meta Llama 3 Community License. SciKnowEval is released under the MIT license. Evaluation benchmarks (AIME, HMMT, AMO-Bench) consist of publicly available math competition problems with no restrictive license.

All artifacts are used consistently with their intended purpose. The DAPO dataset is designed for training mathematical reasoning models. Qwen3-4B-Base and Llama-3-8B are general-purpose language models released for research and development. SciKnowEval is an evaluation benchmark designed for assessing scientific reasoning. We do not release new datasets or models in this submission; our contribution is an algorithmic method applicable to existing RL training pipelines.

Table 4: Hyperparameters used for Math Reasoning Experiments

Parameters	**Value**
General	
Model	Qwen3-4B-Base
Data	
Max. prompt length	2048
Max. response length	8192
Batching	
Question batch size	64
Number of rollouts	16
Rollout	
Inference engine	vllm
Temperature	1.0
Validation	
Temperature	0.6
Top-p	0.95
Number of rollouts	256
Loss	
ϵ	0.2
Rollout importance sampling clip	2
Training	
Optimizer	AdamW
Learning rate	1×10^{-6}
Warmup steps	10
Weight decay	0.01
Gradient clip norm	1.0

Table 5: Hyperparameters used for Scientific Reasoning Experiments

Parameters	Value
General	
Model	Llama-3-8B
Data	
Max. prompt length	2048
Max. response length	8192
Batching	
Question batch size	32
Number of rollouts	8
Rollout	
Inference engine	vllm
Temperature	1.0
Validation	
Temperature	0.6
Top-p	0.95
Number of rollouts	16
SDPO	
JSD α	0.5
Teacher privileged info	Correct rollouts
Training	
Optimizer	AdamW
Learning rate	1×10^{-5}
Warmup steps	10
Weight decay	0.01
Gradient clip norm	1.0